Beta-Testing Without Beta-Testers: Agent-Driven Closed-Loop Game-Design Optimisation for Monopoly

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Abstract

Balancing a game traditionally demands many human testers across many sessions, making design iteration slow and expensive. We present a closed-loop, agent-driven workflow that automates the inner loop of playtesting: a genetic algorithm searches a 66-dim Monopoly board-design space against a composite objective (fairness, length, draw rate, money transfer), scoring each candidate against a fixed pool of 30 diverse rule-based strategies, with an instruction-tuned LLM (Qwen2.5-1.5B) as an independent cross-class probe. The optimised board improves the composite by ∼47% over the default at 2 and 3 players, and three results validate it: **(1)** the 2p/3p-trained boards keep a ∼2× advantage at every count from 2 to 8, except fairness, which is regime-specific; **(2)** both evaluator classes prefer the pool-driven board over an LLM-trained one on 4 of 5 metrics, so a diverse pool generalises across agent classes at zero per-decision LLM cost; and **(3)** in a human pilot (N=10, two players) all trained effects survive, with humans showing a *larger* default-board failure than either simulator, so the agent predictions are conservative, not optimistic. Beyond Monopoly, the recipe transfers to any parameterised multi-agent system: express balance as a small measurable objective, score designs against a deliberately diverse pool of cheap imperfect agents (which exposes blind spots a single strong agent hides), and cross-check the winner with an independent agent class before trusting it. Code, logs, and figures: https://github.com/Selim-Emir-Can/CS348K-proj/tree/master.

1 Background and setup

The problem. Game designers face combinatorial design spaces that human playtesting cannot cover. Tabletop Monopoly is a sharp testbed for the agent-assisted alternative: a small but rich parameter space and well-known balance issues (games drag on, single-property luck dominates, two-player play is unbalanced). The goal of this project is not to "solve" Monopoly but to demonstrate a reproducible designer-facing beta-testing loop:

> **Parameterise environment → evaluate against a diverse agent population → compare agreement across agent classes → shortlist candidate designs → send only the most informative cases to human playtest.**

Inputs and outputs. The pipeline takes a board configuration $\theta \in \mathbb{R}^{44} \times \{0,1\}^{22}$ (22 cost multipliers, 22 rent multipliers, 22 keep-mask bits) and produces (i) a scalar composite score plus a per-component metric tuple $(\overline{F}, F_{\max}, \overline{R}, \overline{D}, \overline{T})$ estimated over 100 games against the strategy pool, and (ii) a decision-quality artefact (per-strategy heatmap, per-game JSONL, or per-decision LLM log) that lets the designer audit *why* θ scored as it did.

Goals and constraints. What makes a game of Monopoly *fun*? We take a good game to be one that (i) stays competitive, with every player keeping a plausible path to victory rather than one seat or strategy running away with it; (ii) fits a single sitting, neither dragging on long after the outcome is clear nor ending before choices matter; (iii) reaches a decisive winner instead of stalemating to a time-out; and (iv) keeps money and property actively changing hands, so turns stay tense rather than passive accumulation. **We chose the objective's metrics to make each of these qualities measurable: fairness (mean and**

worst-pair) for competitiveness, length for sitting-time, draw rate for decisiveness, and money transfer for liveliness. Optimising their weighted sum, rather than any single proxy, forces "fun" to be a balance of all four at once. Concretely, the optimiser minimises a weighted sum of five design qualities (Eq. 1), chosen so that no single term dominates: mean pair-fairness $\overline{F}$, worst-pair fairness $F_{\max}$, length deviation $|\overline{R} - R^*|/R^*$, mean draw rate $\overline{D}$, and player-to-player money-transfer deviation $|\overline{T} - T^*|/T^*$. Hard constraints: every candidate runs end-to-end at the per-property cost/rent bounds $[0.5, 2.0]$, retains both utilities and railroads, and finishes or truncates within 200 rounds. Soft constraints: total wall-clock budget under 30 minutes on a single CPU ($\approx$ 6 ms/game) for the rule-based pipeline, and under 6 hours on a single 12 GB GPU for the LLM probe.

$$\mathrm{score}(\theta) \;=\; w_{\mathrm{fair}}\overline{F} \;+\; w_{\mathrm{fmax}}F_{\max} \;+\; w_{\mathrm{len}}\tfrac{|\overline{R}-R^*|}{R^*} \;+\; w_{\mathrm{draw}}\overline{D} \;+\; w_{\mathrm{money}}\tfrac{|\overline{T}-T^*|}{T^*} \tag{1}$$

Defaults: $w = (1,\ 0.5,\ 0.5,\ 0.3,\ 0.3)$, $R^* = 60$ rounds, $T^* = 100$ \$/round. The GA optimises against $T^* = 100$ \$/round at 2p / 3p training counts; for cross-count reporting (Section 4.3) we use the normalised metric $\overline{T}_{\mathrm{pt}} = \overline{T} \,/\, n_{\mathrm{players}}$ with $T^*_{\mathrm{pt}} = 50$, which equals the original training target at 2p and removes the trivial linear scaling of money flow with seats so cross-count comparison is meaningful.

Hypothesis. The falsifiable claim of the project is two-fold: (a) a candidate board's directional ranking under a 30-strategy parametric pool matches its directional ranking under an independent agent class (a 1.5B-parameter LLM with a deterministic per-field state validator); and (b) the same directional ranking also holds in real human play, so the strategy pool acts as a proxy for human behaviour on the design axes the optimiser targets. (b) is the claim required to use agent feedback as a beta-testing surrogate.

The crux. The hard part is the question of *what evaluates a candidate board* without human playtesters. A single trained agent collapses "design quality" to one playstyle's quirks; the honest alternative is a *population* of strategies, which raises the real question: *which strategies, and how many.* This cannot be shortcut by intuition. Even a seasoned player cannot zero-shot a balanced board or fix the default one with plausible hand-tweaks (dropping properties to shorten the lap, raising rents to force bankruptcies): the 66 interacting knobs trade the five objectives off against one another, so a rent hike that shortens games can hand the win to one archetype, and fairness depends on how *many playstyles* fare against each other, which no designer can introspect. Pinning this down requires playing the games out, which is the point of the project: a population of agents as the evaluator surfaces design insight that intuition and trivial edits cannot.

2 Related work

Automated playtesting and agent-based design evaluation. Procedural and search-based content generation has a long literature [8], and authoring tools such as Sentient Sketchbook [5] use search agents to surface design issues. Closest to our work is the line on *procedural personas* [3], which represents distinct player types as evolved or trained agents and uses them both to playtest and to drive content generation; optimising a design against such an agent population is established prior art. Small instruction-tuned LLMs such as Qwen2.5 [7] have likewise served as decision-making agents, and generative agents [6] treat LLMs as populations of social actors. Our contribution is not any single evaluator but a *protocol that cross-checks multiple agent classes*: a fixed parametric rule-based pool and an independent LLM probe (a single 1.5B model with greedy decoding, a structured `STATE/ECHO/REASON/ANSWER` prompt, and a deterministic per-field validator; failure modes in Appendix A.3), with disagreement-as-diagnostic and human playtesting reserved for boundary cases. The pool uses fixed hand-specified and sampled rule-based parameters rather than trained policies, since per-candidate model-free RL is impractical for a high-variance stochastic game evaluated over thousands of designs.

Tree search and evolutionary search. Browne et al. [1] survey Monte Carlo tree search, and population-based competitive evaluation has been used at scale for StarCraft II leagues [10]; we reuse the "population as evaluator" intuition in inverted form, fixing a diverse population and searching over the environment instead. The search itself is a standard GA [2] used unchanged.

3 Approach

The workflow is built around the game designer: they specify the objective weights and targets (Eq. 1) and the strategy pool, the GA proposes board designs, the simulator scores each candidate against the pool, and the cross-class checks together with the playtest pilot indicate which designs are worth taking to real users (Figure 1). The system is implemented on top of a single-process Monopoly core game loop and pretrained Qwen2.5-1.5B-Instruct [7] weights; component-level details are given below and in Appendix A.

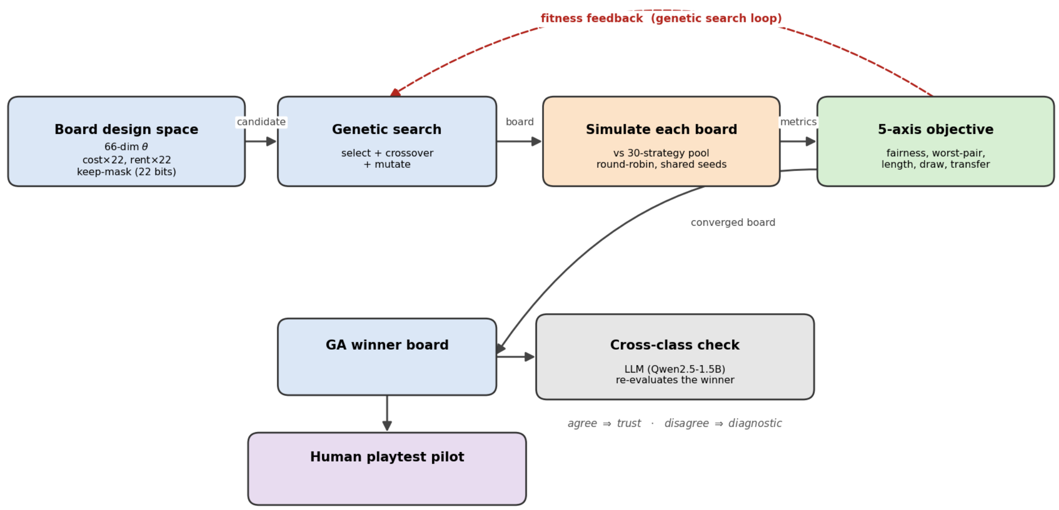


Figure 1: **Closed-loop game-design optimisation.** The GA proposes a board vector θ; the simulator evaluates each candidate against a fixed 30-strategy rule-based pool at shared seeds; the composite score and per-component metrics are returned to the GA. The same winning board is then re-scored under an independent agent class (all-LLM seats with structured echo validation) and against real human play in a small pilot. The designer controls the objective weights and the pool composition; everything downstream is deterministic given a seed.

The implementation is laid out as a standard optimiser/simulator split. The rule-based stack is a 17-knob parametric player, a 66-dim design-space encoder, a thin in-loop simulator that wraps the upstream Monopoly core game loop (built on the PettingZoo multi-agent API [9]) to return per-game stat dicts, and a GA with a matched-budget random-search baseline. The LLM layer is a single player subclass with the structured `STATE/ECHO/REASON/ANSWER` prompt and per-field echo validator described in Section 1. A module-by-module breakdown and the canonical driver scripts are in Appendix A.1; the LLMPlayer prompt and validator are spelled out in Appendix A.2.

Two design decisions did most of the work for the evaluation sections that follow. First, shared random numbers across candidates make GA convergence visible at 100 games per candidate rather than the ~ 1000 that uncorrelated evaluations would require: every candidate is scored on the same 10 matchups $\times$ 10 seeds, and an unchanged design vector yields byte-identical metrics. Second, the 30-strategy pool is the evaluator: a single agent would collapse design quality to its quirks; the pool makes worst-case fairness a first-class objective.

What did not work. Two iterations are worth recording. Qwen2.5-0.5B-Instruct as the LLM probe was uninformative; it emitted near-uniform `BUY` regardless of cash or opponent-completion state, so smaller-than-1.5B parameters did not clear the bar for the cross-class probe to be meaningful. The v1 prompt with a free-form validator induced retry-fabrication (47 retry-induced hallucinations across 2,304 calls) before we realised the issue was a type-coerced cash field in the validator, not the model; v2's structured ECHO block plus deterministic per-field validation drives this to zero across 2,207 calls (Appendix A.3).

4 Evaluation and results

Definition of success. The project is successful if all four of the following hold:

1. the GA finds a board that improves the composite score over the default by a margin larger than the noise floor, verified at the optimum with n=1000 and 95% Wilson intervals on every metric;
2. single-objective ablations show that each of the five composite terms drives a different optimum, establishing that the objective is non-redundant;
3. an independent agent class (LLM under structured ECHO validation) reproduces the same directional ranking on the optimised boards, at a per-decision hallucination rate below 1%;
4. the predicted effects survive contact with real human play, in the same *direction* and similar *relative magnitude* as under the agent evaluators (the bar is direction + relative magnitude across evaluator classes, not exact win-rate matching).

A fifth claim, treated as diagnostic rather than headline, is that running the GA with the LLM as evaluator should converge and beat the default under either evaluator class; the head-to-head verdict between the resulting LLM-driven board and the rule-based GA winner is then reported as a separate finding (Section 4.6), not a precondition for success.

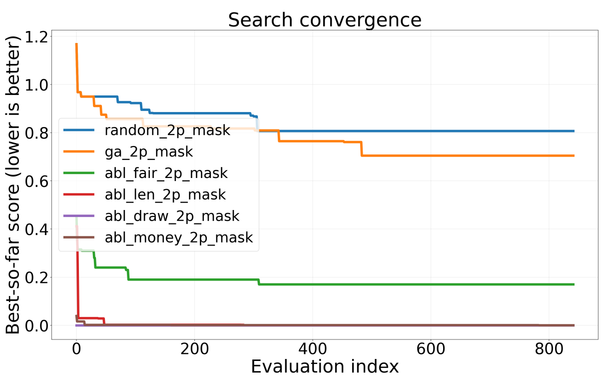
Experimental setup. The strategy pool is 30 parametric players (10 archetypes plus 20 sampled), pickled once with a fixed seed. For each player count $n \in \{2, 3\}$ we draw 10 evaluation matchups under diversity constraints: every archetype appears in at least one matchup, and at least 3 matchups include a sampled strategy. Each candidate is scored on $10 \times 10 = 100$ games with seeds shared across all candidates, so an unchanged design vector yields byte-identical metrics. Hardware: a single CPU at ≈ 6 ms/game for the rule-based stack; a single 11 GB GPU locally for LLM games at $\approx$ 6–10 s/decision. Total games for the entire study: $\approx$ 165k for the rule-based pipeline, $\approx$ 2300 for the LLM cross-class probe, and $\approx$ 2000 for the LLM-driven GA and its cross-evaluation.

4.1 Default-board baseline and search

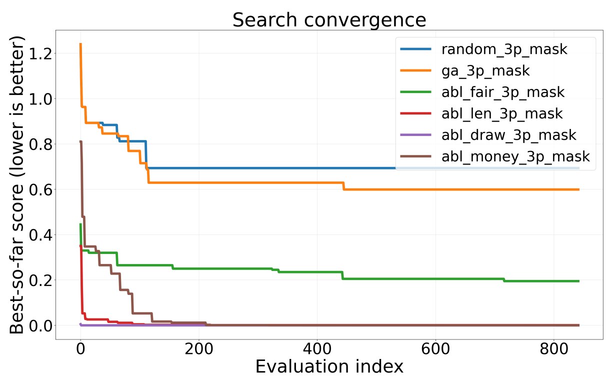
The default board is the identity vector under the encoder. It scores 1.463 on the composite at 2p and 1.230 at 3p (full metrics in the `identity_default` rows of `logs/optimizer_v3/cross_eval_mask.json`); these are the reference points every optimisation result is compared against, with Wilson 95% CIs of $\pm$3pp on individual win rates at the n=1000 deployment evaluation.

The GA outperforms random search consistently across the run, but not by an order of magnitude (Figure 2). The larger lift comes from problem framing rather than optimiser cleverness: the composite objective, the fixed strategy pool, and shared random numbers across candidates make random search on this problem already a strong baseline. This is consistent with a more general observation about system-design problems, where the bottleneck is usually problem formulation rather than search.

What does the GA actually build? Figure 3 renders the combined-objective GA winners alongside the default board for each player count. The 2p winner shrinks the lap from 40 cells to 30 (10 properties removed by the keep-mask), retains both railroads and utilities, and rebalances the remaining colour-group costs and rents. The 3p winner shrinks differently and keeps a different subset of properties, reflecting that a 3-player game pressure-tests a different strategy distribution and converges to a different physical board.

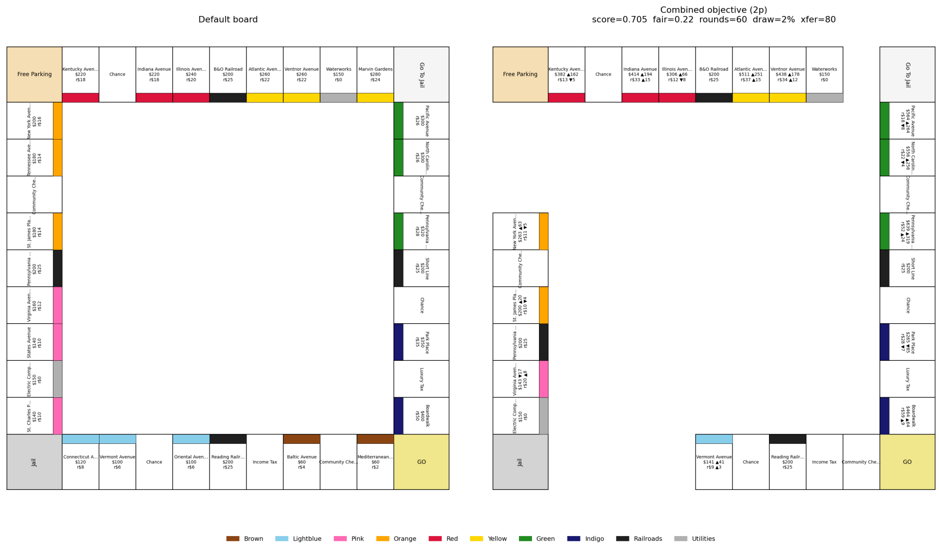


(a) 2 players

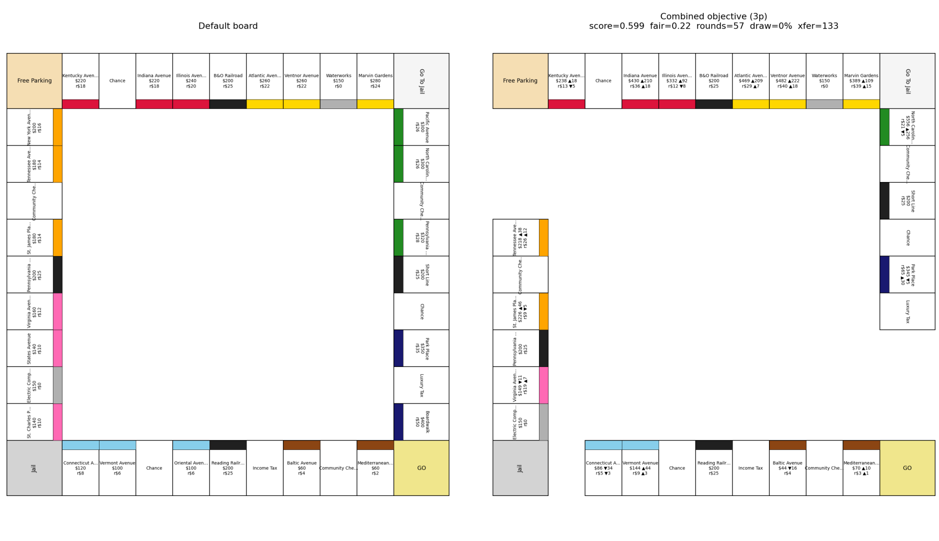


(b) 3 players

Figure 2: **The GA outperforms random search at matched budget by ∼13% in both player counts.** Best-so-far composite score (lower is better) over the candidate-evaluation budget: random search (842 iters), the combined-objective GA (842 evaluations), and the four single-objective ablations. At 2p the GA finishes at 0.705 vs. random's 0.807; at 3p 0.599 vs. 0.694. Ablation runs reach much lower scores because each minimises only one composite term (Figure 4). The lift over the default board (∼47%) is several times the GA-vs-random gap: problem framing dominates, optimiser choice contributes a smaller margin on top.



(a) 2-player combined-objective GA winner.

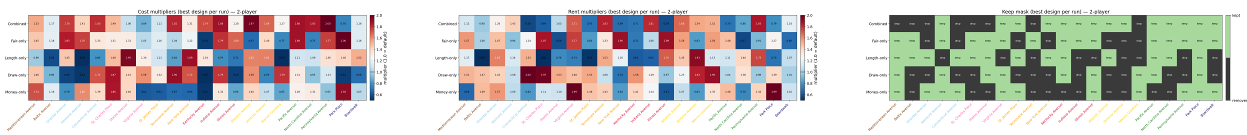


(b) 3-player combined-objective GA winner.

Figure 3: **The boards the GA actually builds.** Each panel shows the default Monopoly board (left, 40 cells) alongside the GA winner (right, shrunk by the keep-mask), with per-property cost/rent multipliers and keep/drop annotations on the optimised side. The 2p and 3p winners are visibly different physical boards, consistent with the cross-evaluation finding (Section 4.3) that each design specialises to its training regime.

4.2 Single-objective ablations

We ran five ablations per player count, each setting one weight to 1 and the others to 0 (*fairness-only, length-only, draw-only, money-only, combined*), and reported best-so-far convergence and the full metric tuple at the optimum. Each ablation reaches a different optimum: the length-only run drives $\overline{R}$ to within ≈ 1 round of $R^* = 60$ but degrades fairness, while the fairness-only run nearly halves $\overline{F}$ but inflates draws. The per-aspect heatmaps in Figure 4 show the ablations carving visibly different shapes in the cost-multiplier, rent-multiplier, and keep-mask sub-spaces. The composite is therefore non-redundant: if any two terms drove the optimiser toward the same board, the corresponding ablations would have near-identical metrics, which they do not.



(a) cost mult. (2p) (b) rent mult. (2p) (c) keep-mask (2p)

Figure 4: **Each ablation carves a different shape in design space (2p).** Per-property cost multiplier (a), rent multiplier (b), and keep-mask bit (c) at the optimum of each GA run. Within each panel, *rows* are the five ablations (*combined, fair-only, length-only, draw-only, money-only*, top to bottom) and *columns* are the 22 ownable properties in board order (Mediterranean Avenue at left through Boardwalk at right). Cell values are the multiplier at the optimum (cost/rent panels, 0.5 to 2.0, with 1.0 being the default identity) or the keep-bit (keep-mask panel: green = kept, black = dropped). If any two ablations drove the optimiser to the same board, those rows would be near-identical; the visible inter-row variation is the empirical evidence that the composite objective is non-redundant.

The same finding is visible at the level of physical boards (Figure 15 in Appendix A.8): the five winners (combined + four ablations) are five visibly different shapes, with different properties kept or dropped and different costs/rents.

4.3 Cross-player-count evaluation (2–8 players)

We cross-evaluate the 2p- and 3p-trained GA winners across all seven player counts from 2 to 8 at n=1000 games per cell, balanced across cyclic seat rotations (Table 1, Figure 5). For cross-count comparison we report two normalised metrics that are described in Section 3 but worth restating here: *transfer per player-turn* = (\$/round) / n_{players}, with target 50 (the original 100 \$/round target evaluated at the 2p training regime); and the *cross-count-comparable composite*, computed with the same weights as the training objective (Eq. 1) but with the transfer term against this per-player-turn metric.

The headline finding is that *game length, decisiveness, money transfer, and the composite advantage all generalise from the 2p/3p training regime to every player count from* 2 *to* 8*, while fairness is strongly dependent on the number of players* (Figure 5). The optimised boards stay near the target 60 rounds at every count (GA-2p winner: 62–68 rounds; GA-3p winner: 55–57 rounds) while the default climbs from 104 rounds at 2p to 158 at 8p. Decisiveness mirrors this: the default truncates 7% of games at 2p, climbing to 71% at 8p (at 8 players nearly three games in four do not resolve within the 200-turn cap); the GA winners stay at most 22% at 8p (GA-3p winner only 12%). On the cross-count-comparable composite, both GA winners maintain roughly a 2× advantage over the default at every count: default 1.29–1.74 across 2–8 players, GA winners 0.51–0.90 across the same range. Money transfer per player-turn rises modestly across counts in the optimised boards (37→57 for the GA-2p winner), passing through the target 50 around 4–5 players, while the default plateaus near 32 \$/player-turn from 3p onward (far below target).

Fairness is the one axis that does not transfer cleanly. Within the training regimes (2p, 3p), the GA winners are 30–53% fairer than the default board (relative mean fairness $\overline{F}$). From 4p onward, that advantage erodes and then inverts: at $7-8p$ the GA winners are $69-101\%$ *less* fair than the default by the same metric. Part of this is metric: $\overline{F}$ is a win-rate spread that mechanically compresses toward zero as seats grow (baseline win rate is $1/n$), so the default's apparent improvement at high counts is largely an artefact

Design	Eval	score ↓	$\overline{F}$ ↓	$F_{\max}$ ↓	rounds (→ 60)	draw % ↓	transfer/turn (→ 50)
default board	2p	1.463	0.454	0.940	103.9	7.4	24.8
default board	3p	1.329	0.412	0.680	110.8	17.6	33.1
default board	4p	1.385	**0.309**	0.640	125.6	38.3	34.3
default board	5p	1.434	0.269	0.680	131.0	46.9	34.6
default board	6p	1.292	**0.181**	**0.340**	140.0	57.0	32.8
default board	7p	1.376	**0.152**	**0.330**	150.7	65.0	31.9
default board	8p	1.401	**0.117**	**0.290**	157.6	71.4	31.4
GA-2p winner	2p	**0.774**	**0.215**	**0.910**	**62.2**	1.7	**36.7**
GA-2p winner	3p	0.729	**0.273**	0.730	65.7	3.3	**44.4**
GA-2p winner	4p	**0.646**	0.310	0.580	63.4	3.9	**49.0**
GA-2p winner	5p	**0.494**	**0.235**	**0.420**	**62.7**	5.4	51.8
GA-2p winner	6p	0.535	0.242	0.420	**64.3**	8.1	**53.8**
GA-2p winner	7p	0.572	0.237	0.400	66.8	14.3	**55.8**
GA-2p winner	8p	0.548	0.198	0.360	67.5	21.6	**57.1**
GA-3p winner	2p	0.897	0.287	0.970	56.2	**0.5**	34.7
GA-3p winner	3p	**0.602**	0.280	**0.490**	**56.3**	**0.6**	42.6
GA-3p winner	4p	0.656	0.336	**0.550**	**56.7**	**1.8**	48.0
GA-3p winner	5p	0.603	0.314	0.460	54.8	**1.8**	**51.7**
GA-3p winner	6p	**0.522**	0.264	0.360	55.4	**4.4**	54.5
GA-3p winner	7p	**0.559**	0.281	0.360	**55.8**	**7.0**	57.1
GA-3p winner	8p	**0.514**	0.235	0.320	**56.7**	**12.4**	59.1

Table 1: **Cross-player-count evaluation at** n=1000 **games per cell across 2–8 players,** 66**-dim keep-mask design space.** All seat permutations balanced via cyclic rotations (extended to n = 4–8). Header arrows: ↓ means lower is better; (→ T) means the target value is T and deviation in either direction is penalised. **Bold** marks the best (lowest) value in each column among the three designs at that player count; underline marks the second-best. The **score** column is the cross-count-comparable composite (transfer term against $/player-turn vs. target 50); **transfer/turn** = ($/round) / n_{players} (target 50), removing the trivial linear scaling of money flow with seats.

of the measure, not the design. A second component is genuine: the GA tuned structural advantages for the training matchups, and seating additional pool strategies amplifies those advantages into wider win-rate spreads. The takeaway is that fairness is *strongly dependent on the number of players*: it is the one metric that does not transfer cleanly from the 2p/3p training regime to higher counts, and the GA's fairness gains do not generalise the same way pacing, money-circulation, and decisiveness do.

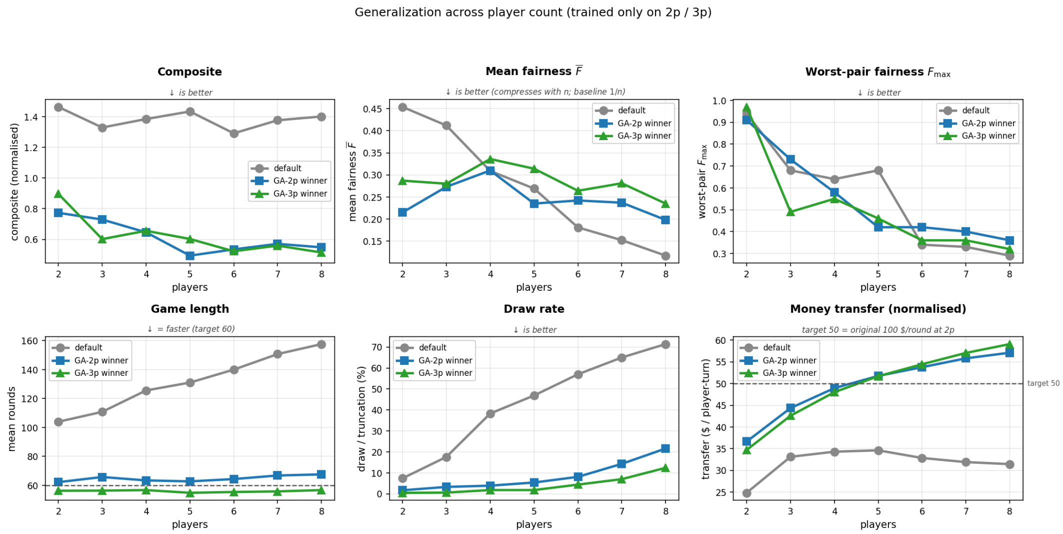


Figure 5: **Generalisation across player count, trained only on 2*p*/3*p*.** All six evaluation metrics from Table 1 across 2–8 players, for the default board (gray) and the two combined-objective GA winners (blue, green). Top row: composite (cross-count-comparable, ↓ better), mean fairness $\overline{F}$ (↓ better; mechanically compresses toward 0 with n because baseline per-seat win rate is $1/n$), worst-pair fairness $F_{\max}$ (↓ better). Bottom row: mean rounds (↓ = faster; target 60), draw / truncation rate (↓ better), and inter-player money transfer per player-turn (target 50 $/player-turn). The optimised boards stay near target rounds and below 25% draw rate at every count, and maintain a clean composite advantage across 2–8 players; fairness $\overline{F}$ is the one axis that does not transfer cleanly (the winners are fairer than default in-distribution and less fair at high counts; part of the apparent default-improvement is the metric compression noted above).

4.4 Archetype reshuffle: *who* wins differently?

The composite score averages over only the 10 sampled evaluation matchups; it can move down even if a different subset of strategy pairs got worse. To diagnose *which* archetypes the GA winner helps and which it hurts, we compute the *win-rate vs. the field* for each pool strategy i as the mean over $j \neq i$ of the underlying 30×30 pairwise win-rate matrix $W^{\text{board}}[i, j]$ (20 games per ordered pair, 580 games/archetype/board). Table 2 reports the 10 named archetypes individually, sorted by $\Delta = \text{WR}_{\text{GA-2p}} - \text{WR}_{\text{default}}$, with the 20 sampled strategies aggregated as a single row.

Archetype	WR(default)	WR(GA-2p)	Δ	Identity
CashHoarder	0.46	0.63	+**0.18**	$1k+ unspendable, no trades
RiskAverse	0.50	0.60	+0.10	hoard, skip expensive groups
Passive	0.48	0.54	+0.07	no trades, no rails/utils
LowCostOnly	0.56	0.62	+0.06	skip 5 expensive groups
Bully	0.63	0.62	−0.00	ultra-aggressive build + trade
RailroadKing	0.01	0.00	−0.01	rails only, ignore colour groups
Trader	0.66	0.60	−0.05	aggressive build + trade
AggressiveBuilder	**0.73**	**0.66**	−0.06	build at zero floor, no trades
Balanced	0.71	0.64	−0.06	balanced thresholds, trades
HighCostOnly	0.56	0.38	−**0.18**	skip 3 cheap groups
Random sampled (n=20)	0.45	0.48	+0.03	20 random draws from the parameter space

Table 2: **Per-archetype win-rate against the field, default vs. GA-2p winner.** WR vs. field = mean win-rate against the other 29 pool strategies, 20 games per ordered pair, 580 games/archetype/board. Sorted by Δ descending. Largest positive and negative deltas in bold.

The board change does *not* flatten win-rates uniformly; it reshuffles which archetypes are top-tier. Slow / cash-conserving archetypes (CashHoarder +0.18, RiskAverse +0.10, Passive +0.07, LowCostOnly +0.06) move from middling to top-tier on the GA winner, and HighCostOnly collapses (−0.18) because the keep-mask prunes the expensive groups it relies on. The mechanism is the same *expansion-to-solvency* flip the human playtest exposes (Section 4.7): on the default board (∼100 rounds, low rents) the optimal play is to own more, so AggressiveBuilder dominates; on the GA winner (∼60 rounds, ∼1.3× rent, ∼1.6× cost) one bad landing on a developed cell can drain a $300 reserve, so cash-discipline becomes the dominant trait. AggressiveBuilder's WR against CashHoarder collapses from 0.80 to 0.45 across this single board change.

A second observation, harder to see from the means, is that the reshuffle is concentrated in the middle of the win-rate distribution. The top-3 default-board strategies (AggressiveBuilder, Balanced, Trader, all with WR ≥ 0.66 on default) lose only -0.05 to -0.06 on the GA winner; they remain top-3 on both boards. At the other extreme, RailroadKing is pinned near zero on both boards because it ignores every colour group and any opponent who builds wins, regardless of board parameters. The top and bottom of the WR distribution are board-robust under our search space; the middle is where the GA's reshuffle lives. This is consistent with the GA winner being a mutation of the default (per-property cost / rent / keep-mask) rather than a board synthesised from scratch. The takeaway is that archetype performance is a property of the *(strategy, board)* pair, not of the strategy alone: a single agent cannot tell us whether a board is well-designed, and a single board cannot tell us which strategies are well-designed.

A related observation, from the underlying pairwise matrix, is that some matchups are immutable under any board configuration in the search space. The most asymmetric pair on the GA-2p winner remains *Trader* vs. *RailroadKing* (100%/0%). No setting of cost multipliers, rent multipliers, or keep-mask bits eliminates this; it is intrinsic to a strategy that refuses to buy 22 of 28 ownable properties, and the GA cannot reach it through the design space. Board design is a strong lever for system-level properties (length, draws, money transfer) and for matchup-set fairness, but agent-level dominance on specific strategy pairs is structural to the strategy pool. Closing those residual gaps requires a strategy-side intervention (rule changes, matchmaking, or pool curation), not more search over the design space.

4.5 LLM cross-class agreement

The cross-class probe re-runs the optimised boards under all-LLM seats (Qwen2.5-1.5B-Instruct, greedy decoding) with the structured `STATE/ECHO/REASON/ANSWER` prompt; per-decision logs record prompt, raw response, parse path, and per-field echo mismatches. Across 2,207 LLM calls under the v2 prompt, the probe produced zero model-side state hallucinations under deterministic echo validation. The full v1-vs-v2 hallucination audit (including a second-order finding about retry-induced confabulation under a buggy validator) is in Appendix A.3.

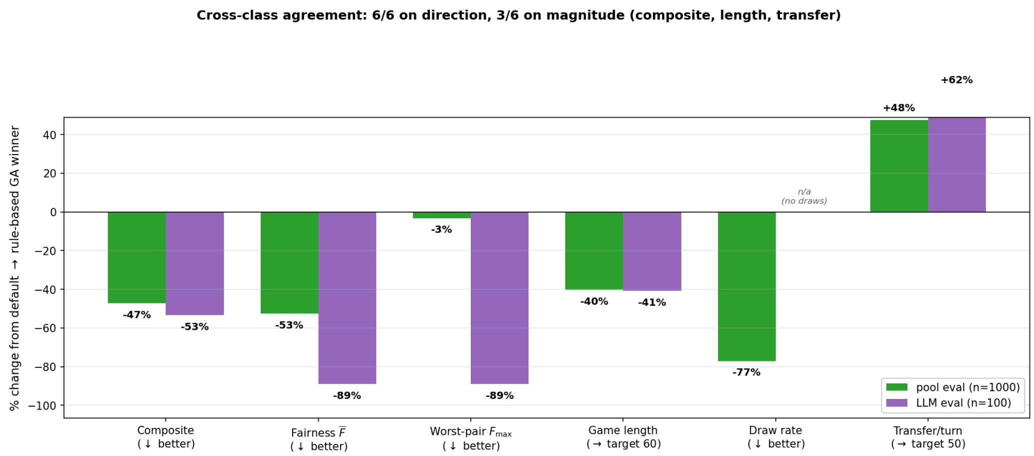


Figure 6: **Both evaluator classes agree on direction for all 6 metrics (rule-based GA winner vs. default, 2-player).** Per-metric % change under the 30-strategy pool (green, n=1000) and all-LLM seats (purple, n=100). Magnitudes match closely on composite ($-47\%/-53\%$), length ($-40\%/-41\%$), and transfer per player-turn ($+48\%/+62\%$); the two fairness terms agree in direction but diverge in magnitude (smaller LLM tail at n=100). Draw rate is in the panel footer because LLM seats produce no draws on either board.

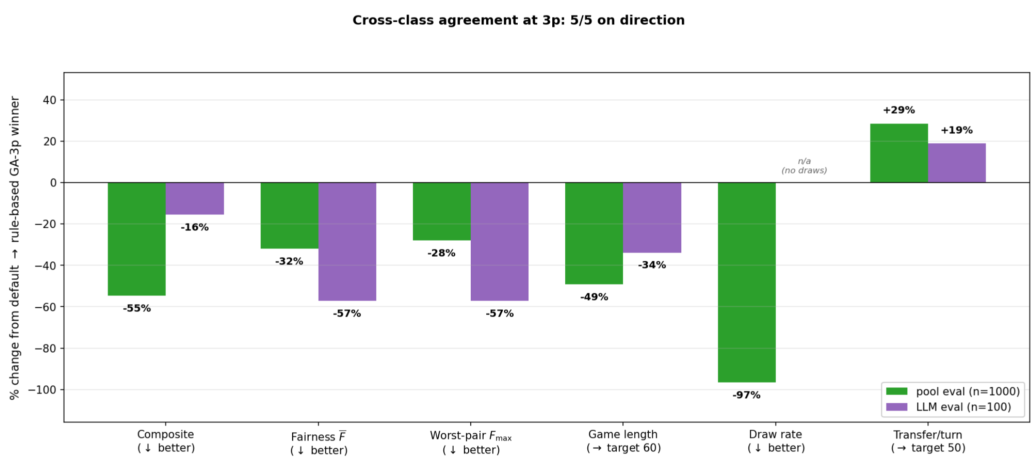


Figure 7: **Cross-class agreement at 3p: same-direction improvement on every measured metric.** % change default → GA-3p winner under pool (green, n=1000) and LLM seats (purple, n=100). Direction agrees on 5/5; magnitudes match on length ($-49\%/-34\%$) and transfer per player-turn ($+29\%/+19\%$). The 3p audit reproduces the 2p result (Figure 6).

4.6 LLM-driven GA and the cross-class verdict

We re-ran the GA with the LLM as evaluator (population 8, 5 generations, 5 seeds per candidate, 2-player only). The LLM-driven loop converges and produces a winning board that beats the default under all-LLM evaluation (the convergence curves and per-generation score distribution are in Appendix A.6). Cross-evaluating that board against the rule-based GA winner under both agent classes (Figure 8) produces the central finding of the project: both evaluator classes prefer the rule-based GA winner on 4 of 5 stable metrics (composite, $\overline{F}$, $F_{\max}$, game length). The LLM-driven board wins on only 1 metric, transfer per player-turn, and that is the metric its small-n training overfit to. On the four metrics that drive the composite, the strategy-pool-driven design generalises and the LLM-driven design does not.

The interpretation is the headline finding of the project: a sufficiently large rule-based strategy pool

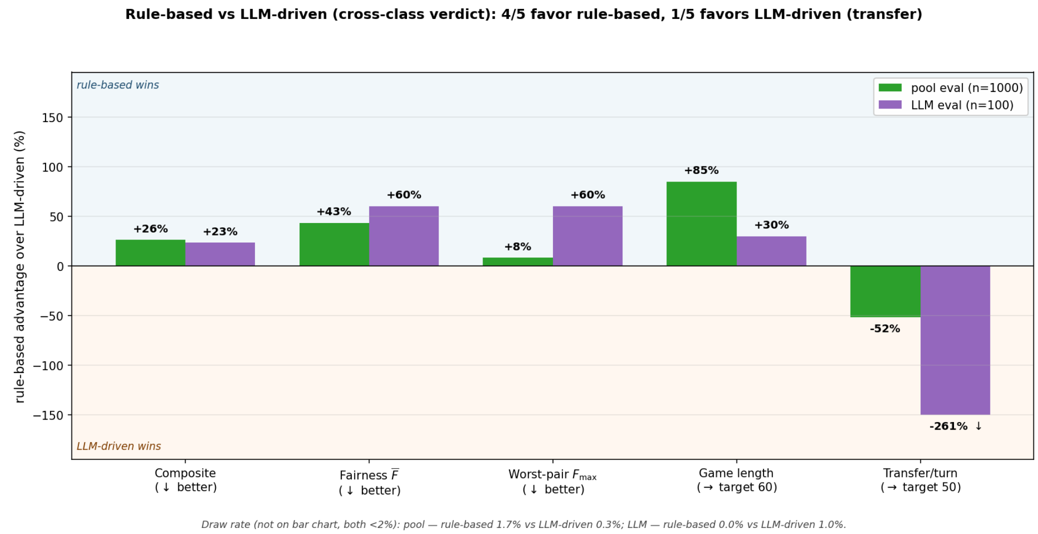


Figure 8: **Both evaluator classes deliver the same verdict: rule-based wins on 4/5 metrics, losing only on transfer.** Per-metric % advantage of the rule-based GA winner over the LLM-driven board (2-player) under pool (green, n=1000) and LLM seats (purple, n=100); positive places the bar in the rule-based-wins zone (blue), negative in the LLM-driven-wins zone (orange). Both evaluators put 4/5 bars in the rule-based zone (composite, both fairness terms, length); the LLM-driven board wins only on transfer per player-turn, the metric the LLM-GA's n=5 training overfit to. Draw rate is in the footer (both $<2\%$). LLM-eval transfer bar clamped at -150% for comparability (true value -261%, marked ↓).

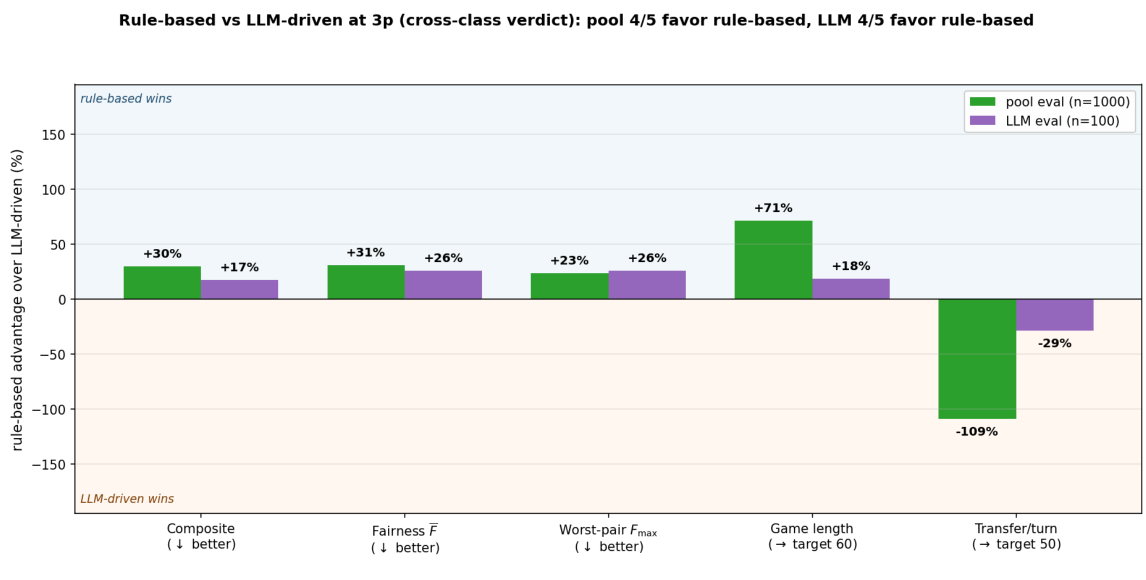


Figure 9: **Cross-class verdict at 3p: rule-based wins 4/5 under both evaluators, same pattern as 2p.** Per-metric % advantage of the rule-based GA-3p winner over the LLM-driven board (3-player, n=100 LLM seats). The same 4/5 metrics that favour rule-based at 2p also favour it at 3p; the one disagreement (transfer/turn) is again the LLM-GA's small-n overfitting target. Draw rates in footer (both $< 1\%$).

produces an optimised design that generalises to an evaluator class never seen at training (LLM seats), at zero per-decision LLM cost. The pool's 30 strategies expose the GA to many play styles, so it cannot overfit to any single one; the LLM evaluator's narrow decision distribution makes its own optimiser overfit, which is why the pool-driven board beats the LLM-driven board on 4/5 stable metrics under both evaluators and loses only on the metric the LLM-GA's small-n training overfit to. A single LLM or RL agent as evaluator is a benchmark; a diverse strategy pool is a beta-testing surrogate. The 3p audits (Figures 7, 9) confirm this verdict holds when the GA winner is itself a 3-player design, not just when the rule-based winner is the 2p board.

4.7 Human playtest pilot (N=10, two players)

As a reality check on whether the agent-optimized boards behave as predicted under human play, two players played two two-player boards: the default and the GA-2p combined winner (Section 4.1). Each board was played ten times under matched RNG seeds (0, 1, 2, 3, 4, 5, 6, 7, 8, and 42), a common-random-numbers pairing across boards so that a difference reflects the board rather than luck. The four per-game observables the search optimizes that are legible from the game transcripts, game length, inter-player money transfer, draw / truncation rate, and the per-player win rate, were recovered with `scripts/human_game_stats.py`.

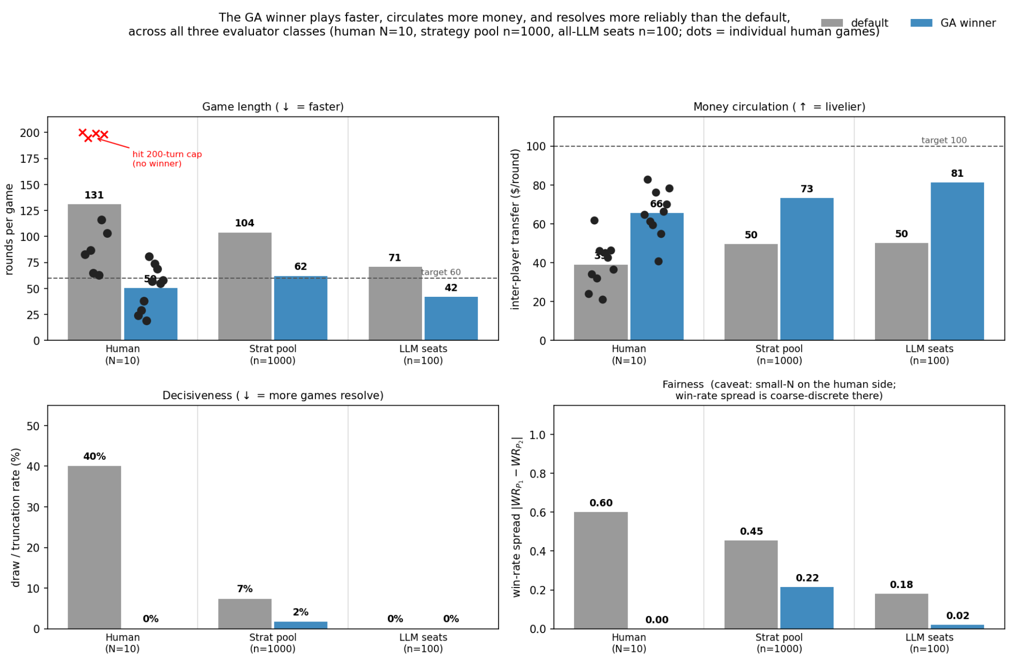


Figure 10: **The GA winner plays faster, circulates more money, resolves more reliably, and is perfectly balanced under human control.** Per-board mean over ten games at matched seeds; dots are individual games. **Top-left:** GA winner 50 rounds vs. 131 default; four default games hit the 200-turn cap (red crosses). **Top-right:** \$66/round vs. \$39/round (target \$100). **Bottom-left:** default truncates 4/10, GA winner 0/10. **Bottom-right:** win-rate spread 0.60 default vs. **0.00** GA winner (5–5 split). N=10 per board, two players.

Figure 10 summarizes the twenty games. The GA winner finished in 50 rounds on average against 131 for the default ($\sim 2.6\times$ faster), and four of the ten default games failed to resolve at all, hitting the 200-turn cap with no winner. All ten GA-winner games ended decisively by bankruptcy. Inter-player money transfer per round was $\sim$ 68% higher on the GA winner (\$66/r vs. \$39/r), and the win-rate spread was 0.00 on the GA winner (a perfect 5–5 split between the two players) against 0.60 on the default (Player 1 won all six decisive games).

Success criterion. We frame the bar this pilot clears explicitly. The agent-driven tool succeeds if two conditions hold under real human play. First, the design constraints specified in the agent setting (shorter games, more decisive endings, more money circulation, fairer 2-player matchups) are satisfied in real play, in the same *direction* and similar *relative magnitude* as under the agent evaluators (rule-based pool, all-LLM seats). Second, the agent feedback is *conservative* rather than optimistic: humans should not exhibit a *smaller* default-board failure than the simulators predicted. Both hold on the four observables we can extract from the transcripts. Direction agrees on every metric across pool / LLM / human; the magnitudes are similar ($\sim$104 vs. 131 default rounds, $\sim$62 vs. 50 GA-winner rounds). On the conservatism bar specifically, the humans showed a *larger* default-board failure than either simulator: a 40% draw rate (4 of 10 default games hit the 200-turn cap) versus 7% in the strategy-pool eval and 0% under the all-LLM evaluator. The agent-driven predictions therefore under-state how broken the default board is for humans, not the other way around.

We treat the fairness spread of 0.00 on the GA winner at N=10 as the strongest possible outcome at this sample size, but acknowledge that a larger study across many subjects would still be needed to rule out single-pair idiosyncrasy in the strategic style. The pacing, money-circulation, and decisiveness effects are large enough that they are unlikely to be artefacts of either subject or seed. The fuller multi-subject playtest that would substantiate these effects across the broader population remains deferred; we specify it next.

Qualitative play-experience differences. The figure summarises four objective metrics, but the matched-seed pairing also exposes how the board's parameter changes translate into a qualitatively different play experience that the metrics alone do not fully capture. On the default board, completing a colour set is slow: each three-property group requires multiple landings, the opponent can actively block by buying mid-group properties to deny a monopoly, and even once a set is completed, immediately building houses is not always optimal (the cheaper side-of-board sets benefit from staged builds and a cash reserve in case the opponent lands elsewhere). On the GA winner, the keep-mask shrinks several colour groups to one or two cells, so a single landing monopolises a group and the player can begin building houses immediately; combined with the higher cost and rent multipliers, every landing on a built property is a sizeable money transfer, and concentrated-investment plays that would be suicidal on the default (e.g., maxing out houses and then a hotel on a single prime cell such as Virginia Avenue in our recorded game 9 on the GA-winner board) can pay off here. The same shrinking that makes set-completion fast also makes it harder for either player to block the other, so games progress more linearly toward a decisive build phase rather than through a long trading and blocking pre-build.

4.8 Game-design insights

The combination of system-level metric changes (Section 4.3), per-archetype reshuffle (Section 4.4), and qualitative play differences (Section 4.7) supports a small set of concrete game-design insights that go beyond "the optimised board is better":

Expansion $\rightarrow$ solvency: the losing condition flips. The default board (long games, low rents) rewards owning more properties: time amortises any acquisition. The GA winner ($\sim$60 rounds, $\sim$1.3$\times$ rent multipliers, $\sim$1.6$\times$ cost multipliers) rewards staying solvent: one bad landing on a developed cell can drain a \$300 cash reserve, and the lap is too short for slow accumulation to recover. AggressiveBuilder's win-rate against CashHoarder collapses from 0.80 on the default to 0.45 on the GA winner; the same shift shows up across the cash-conservative archetypes (Section 4.4). The mechanism is not "aggression is punished"; it is that the losing condition itself changes from *owning too little* to *going bankrupt on a rent landing*.

Smaller colour groups make all-in builds work. On the default board, the only 2-cell colour groups are Brown (cheapest) and Indigo (most expensive); a Brown monopoly, fully built, cannot generate enough rent to win a game on its own, and any other set-completion requires acquiring three specific cells. On the GA winner, the keep-mask shrinks most colour groups to 1–2 cells, and the higher cost / rent multipliers make those small groups individually viable: a max-houses-plus-hotel play on a single prime cell (e.g. Virginia Avenue in our recorded game 9) pays off rather than being a desperation play.

Blocking opponents stops working. On the default, a player can deny an opponent's colour set by buying one mid-completion cell of it, and the long game leaves time for the denial to compound. On the GA winner, most sets are 1–2 cells, so they complete on the first landing and the denial-purchase lever disappears entirely; the higher property costs also prevent hoarding many properties before crossing GO. This is a board-design effect with no analogue in shorter-rent or higher-rent tuning alone: it requires the keep-mask, and it changes which player classes can apply pressure to whom.

Per-metric mechanisms. The four headline metrics each have a specific board-design cause:

- **Game length** (∼60 vs. ∼100 rounds): a 12-cell lap leaves fewer turns between rent events.
- **Decisiveness** (draw rate 7% → 2% under pool, 40% → 0% under human play): ∼1.6× base cost and ∼1.3× base rent force bankruptcy before the 200-turn cap.
- **Money circulation** ($50 → $73/round under pool): higher base rent on cells landed more often.
- **Fairness at 2p/3p** (spread 0.45 → 0.22 at 2p; −32% at 3p): Yellow dropped entirely, most other groups shrunk to 1–2 cells, and per-property rent multipliers spread across ∼0.6×–1.9× to damp over-powered cells. The combined GA converges on the same recipe the fairness-only ablation (Section 4.2) finds, integrated with the length / money / draw terms.

The fairness mechanism deserves a separate note: the GA does not just shrink groups uniformly; it also tunes individual rent multipliers (range 0.59×–1.87× on the GA-2p winner) so that no single colour group is structurally dominant on the shortened lap. This is the same mechanism the fairness-only ablation produces, so its appearance in the combined winner is not an artefact of overfitting to one objective term; it is the GA finding a single configuration that satisfies length, decisiveness, money, *and* fairness simultaneously.

4.9 Full playtest (deferred)

Beyond the N=10 pilot above, the full multi-subject playtest remains deferred. We publish a shortlist of four boards we would test (default, GA-2p combined winner, GA-3p combined winner, money-only 2p optimum), each contrasting a different design lever against the default, with a proposed pilot protocol (N=8, counterbalanced, Likert plus forced-choice) and the per-board design hypothesis the pilot would falsify; see Appendix A.12 for the full specification.

4.10 Limitations

The main caveats: the 17-dim strategy space is broad but not exhaustive; the N=10 pilot rules out idiosyncrasy in trained-effect direction but not in the broader population; fairness is strongly dependent on the number of players (GA winners 30–53% fairer than default at training counts, 69–101% less fair at 7–8p); Qwen2.5-1.5B is the only LLM in the study; and all optimisation runs on the standard 40-cell US Monopoly layout. Full discussion in Appendix A.11.

4.11 Conclusion

Three independent validations clear the bar set in Section 4: (i) cross-player-count generalisation, the 2p/3p-trained GA winners retain a ∼2× composite advantage at every count from 2 to 8 with the single exception of fairness, which is regime-specific; (ii) cross-class agreement, the strategy-pool-driven board beats an LLM-GA-trained design on 4 of 5 stable metrics under *both* evaluator classes at both 2p and 3p, at zero per-decision LLM cost; and (iii) a human playtest pilot (N=10, two players) that confirms the direction and relative magnitude of every trained effect, with humans showing a *larger* default-board failure than either simulator predicted, so the agent-driven predictions are conservative rather than optimistic.

The underlying mechanism is a flip in the losing condition: the default board (long games, low rents) rewards owning more properties, while the GA winner (∼60 rounds, ∼1.3× rent, ∼1.6× cost) rewards staying solvent on a rent landing. That single shift reshuffles which archetypes win on the GA winner (CashHoarder +0.18 in WR vs. the field, HighCostOnly −0.18), neutralises the defensive-purchase blocking lever, and makes max-build plays on small colour groups dominant rather than suicidal.

The methodological lessons follow directly. A sufficiently diverse rule-based pool can substitute for a more expensive evaluator class while generalising to it; a worst-pair fairness term prevents the optimiser from improving the average at the expense of the tail; per-objective ablations expose composite redundancy;

evaluation context (player count) is itself a design variable; shared random numbers reduce variance at fixed sample size; and the verdict that matters is whether a design's predicted effects survive contact with an evaluator class the optimiser never saw, here checked both with an all-LLM cross-class evaluator and a small human pilot. The recipe (closed-loop GA + diverse evaluator pool + independent cross-class checks) transfers to any parameterised multi-agent system.

5 Author contributions

Selim Emir Can. Project idea, implementation, experiments, figures, and writing; all work in this report is mine.

A Implementation details

This appendix collects implementation details that are useful for reproducibility but would clutter the main text. All source paths are relative to `monopoly/` on the project repository.

A.1 Pipeline components and driver scripts

Optimisation stack. The optimisation stack lives under `monopoly/optimizer/` and `monopoly/scripts/`. `ParametricPlayer` together with `ParametricPlayerSettings` subsumes every existing rule-based subclass under a single 17-knob agent (5 continuous, 4 boolean, 8-bit colour-group mask). `strategy_pool.py` produces the 30-strategy pool described in Section 1, deterministic via a fixed seed and saved once for reuse across runs. `design_space.py` maps a 66-dim vector to a fresh `GameConfig` by applying per-property multipliers to `cost_base`, `cost_house`, `rent_base`, and `rent_house`, and physically removing properties whose keep-bit is zero (structurally shortening the lap rather than substituting `FreeParking`); the identity vector is the default board. `simulate.py` is a thin re-implementation of the inner game loop that returns a per-game stat dict (winner, rounds, bankruptcy map, inter-player transfer total) instead of only logging; inter-player cash flow is captured via a context-manager monkey-patch of `Player.pay_money`, with non-player payments (taxes, fines, salary) excluded, and trade calls are capped at 5 per (player, turn) to break an unbounded loop in upstream code. `search.py` implements both random search and a GA (population 20, 10 generations, tournament selection k=3, uniform crossover, σ=0.1 Gaussian mutation on continuous dims, 1-bit-flip mutation on the keep-mask, elitism 2); both produce identical JSONL.

Driver scripts. The end-to-end pipeline is driven by five scripts. `scripts/optimize_board.py` runs the rule-based search and emits one JSONL line per evaluation. `scripts/cross_eval.py` runs designs through both player-count harnesses at n=1000. `scripts/strategy_heatmap.py` produces 30×30 strategy-vs-strategy win-rate matrices on any chosen design. `scripts/eval_llm_on_boards.py` runs the all-LLM-seat probe with per-decision logging, and `scripts/optimize_board_llm.py` re-runs the GA with the LLM in the evaluator role.

A.2 LLMPlayer prompt and validator

Model and decoding. Qwen2.5-1.5B-Instruct [7] loaded with the `transformers` library, run with greedy decoding (`do_sample=False`), `max_new_tokens=160`, and additional EOS tokens `<|im_end|>` and `<|endoftext|>`. The same model and decoding settings are used for all four runs (LLM-only evaluation and the LLM-driven GA, at 2p and 3p).

Conversation structure. Each LLM buy decision is a six-message conversation: 1 system message, 4 few-shot user/assistant exchanges (2 *BUY* and 2 *PASS* demonstrations), and 1 runtime user message describing the actual decision. The full transcript with all four few-shot examples is checked into the repository at `prompts/llm_player_prompts.txt`.

System prompt (v2, verbatim).

> *You are a strategic Monopoly player who decides whether to buy properties. Your goal is to win by bankrupting opponents through rent. You must NOT mindlessly buy every property: passing is correct when (a) cash is dangerously low, (b) the property is in a group an opponent already dominates and you have no path to monopoly, or (c) the rent return is poor relative to the cost. Equally, buying is correct when the property advances a group you already own most of, or when it denies the group to an opponent who would otherwise complete it.*
>
> *You receive a STATE block with 10 fields. Before reasoning, you MUST emit an ECHO block that copies all 10 STATE fields verbatim (numeric values must equal STATE; string values must equal STATE). The ECHO block is checked deterministically – if it differs from STATE in any field you will be asked to redo the answer. Only after the ECHO block do you write REASON and ANSWER. Do not invent any number or name not present in STATE. A purchase ONLY "completes a monopoly" when* `you_own_in_group + 1 == group_size`*; do not claim completion otherwise.*
>
> *Reply in EXACTLY this format and order:* `ECHO:` *followed by the 10 STATE fields (*`cash`*,* `property`*,* `group`*,* `cost`*,* `base_rent`*,* `you_own_total`*,* `you_own_in_group`*,* `group_size`*,* `opp_own_in_group`*,* `opp_own_total`*); then* `REASON: <one short sentence citing only STATE numbers>;` *then* `ANSWER: BUY` *or* `ANSWER: PASS`*. Do not output anything else.*

v1 vs. v2 prompts. The v1 prompt was free-form: it asked for "BUY" or "PASS" with a one-sentence reason but did not require the ECHO copy of STATE. The v1 validator parsed the free-form response with a regex and was prone to a now-fixed integer/float type mismatch on cash (Section 4.5). v2 added (i) the verbatim ECHO requirement, (ii) deterministic per-field equality checking on the echoed values, and (iii) a corrective retry loop that replays prior assistant turns plus a follow-up user message naming the mismatched fields. Both versions share the same four few-shot examples and the same STATE format.

Echo validator. `LLMPlayer._check_echo` parses the ECHO block with a multiline regex and equality-checks each of the 10 fields against ground-truth state, with a $0.50 tolerance on cash to absorb float drift from rent multipliers (the original fixed-int validator caused 100 spurious flags in v1).

Retry mechanism. On any echo mismatch, the player retries up to `MAX_RETRIES = 4` times. Each retry replays prior assistant turns plus a corrective user message that names the mismatched fields. The final attempt's parsed answer is the decision used by the simulator, even if it still has echo issues; the per-decision JSONL log records every attempt for post-hoc analysis.

Pre-filter. Affordability and cash-floor checks short-circuit the LLM call when the decision is forced (cash < cost forces PASS, etc.). This eliminates 60–80% of would-be calls and keeps the LLM evaluation pipeline within practical wall-clock budgets.

A.3 v1-vs-v2 hallucination audit

The cross-class probe (Section 4.5) ran two prompt versions: v1 (free-form BUY/PASS with a one-sentence reason) and v2 (the structured `STATE/ECHO/REASON/ANSWER` prompt described above). v1's validator had a subtle integer/float type mismatch on the cash field, and retries against a wrong validator induced *more* plausible-looking confabulation rather than less, because the model was being told a correct value was wrong and produced a different "corrected" answer to satisfy the validator. We document this here rather than tuning the validator until it disappears: a protocol whose failure modes are reported alongside its results is auditable; one whose are not is a benchmark trick. v2 fixes the validator and adds the verbatim ECHO copy; across 2,207 calls under v2 the model never fabricates state at first-pass and the validator never falsely flags a correct output.

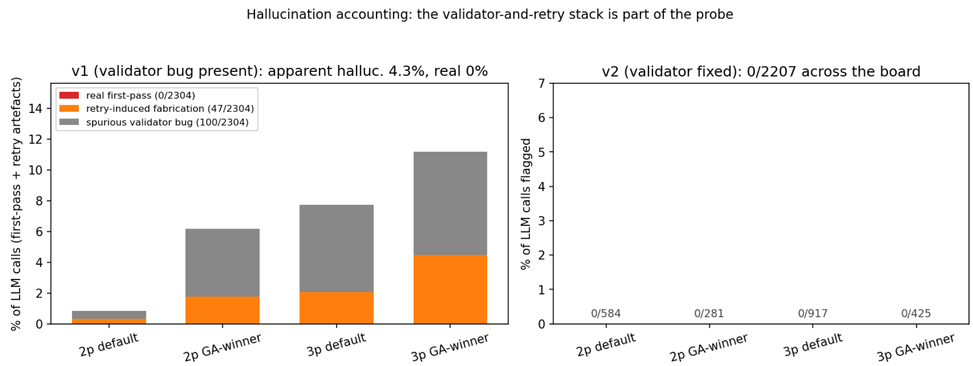


Figure 11: **The validator-and-retry stack is part of the probe: a faulty validator does not just produce noise, it pulls the model toward fabrication.** Per-call validation outcomes for the v1 prompt (left, with a known validator bug) and the v2 prompt (right, bug fixed). Each bar is stacked into three categories: *real first-pass* hallucinations (the model fabricated state on its first attempt), *retry-induced fabrication* (the model's first-pass output was correct but the buggy validator flagged it, and the model produced fabricated state on retry to "satisfy" the flag), and *spurious validator bug* (the validator falsely flagged correct outputs that the model did not change). On v1 across 2,304 calls, real first-pass = 0, retry-induced = 47, spurious = 100; the model itself never hallucinated. v2 drives all three categories to zero across all 2,207 calls. The reportable second-order finding: validating against a wrong ground truth is worse than not validating, because retries can teach the model to produce the "correct" wrong answer.

A.4 LLM buy-rate diagnostics

To verify that the LLM probe is making rational rather than uniform-random decisions, we slice its buy rate by available cash and by monopoly-completion situation on both boards. The probe buys monotonically more often as cash rises, saturates near 1.00 when it already partially owns the colour group, and drops to $\sim$0.13 when the opponent already dominates the group; the directional responses are essentially identical on both boards (Figure 12).

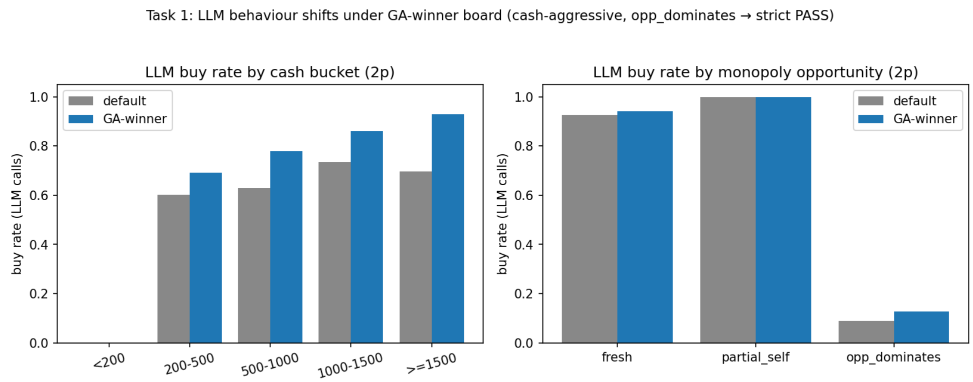
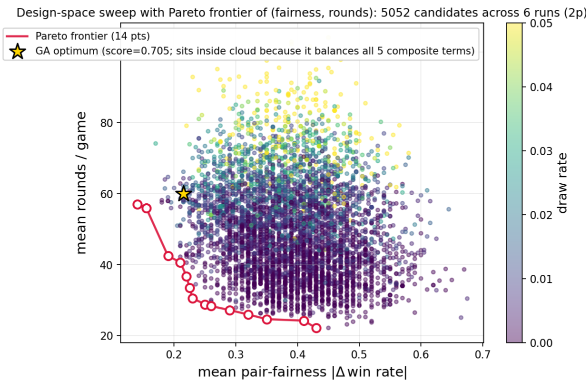


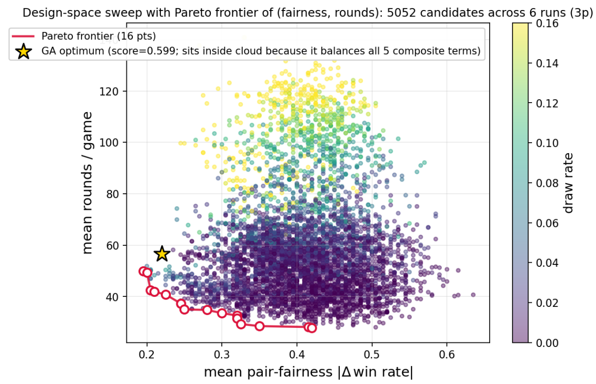
Figure 12: **LLM buy-rate slices, default board vs. GA-2p winner. (a) By cash bucket.** Buy rate conditioned on the LLM's available cash at decision time. The rate rises monotonically with cash, the rational ordering for any non-degenerate strategy; the empty < 200 bucket reflects the affordability pre-filter that short-circuits the LLM call when the player cannot buy anyway. On the GA-winner board the LLM commits its cash hoard more aggressively in the ≥ 1500 bucket (~ 0.93 vs. ~ 0.70 on the default), so the optimised board successfully pulls behaviour out of the cash-hoarding regime. **(b) By monopoly-completion situation.** *fresh*: no player owns any property in the landed colour group, so a purchase starts a new collection. *partial_self*: the LLM already owns one or more properties in the group and the opponent does not dominate it. *opp_dominates*: the opponent owns at least half the group and at least as many properties as the LLM, so buying mostly funds their next rent payment. Buy rates respond rationally across all three: ~ 1.00 on *partial_self* (saturated buy), ~ 0.94 on *fresh*, and ~ 0.13 on *opp_dominates* (strict PASS). The probe is not behaving as a uniform random oracle, and the directional responses are essentially identical on both boards.

A.5 Pareto trade-off surface

The (fairness, rounds) trade-off surface (Figure 13) visualises the design-space geometry that the cross-player-count evaluation (Section 4.3) reads off. The two clouds have visibly distinct shapes, which is what makes that cross-evaluation finding a property of the design space rather than seed noise.



(a) 2p

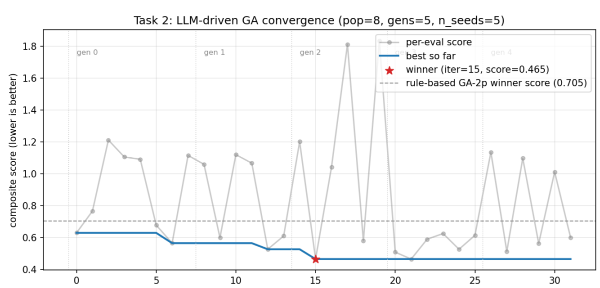


(b) 3p

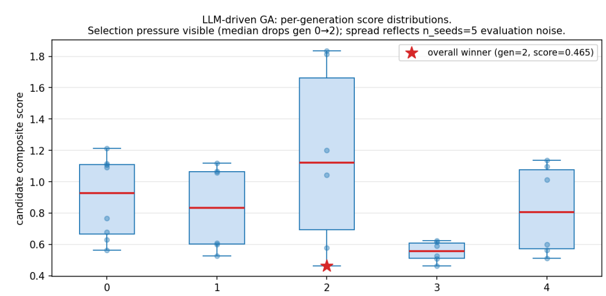
Figure 13: **The (fairness, rounds) trade-off surface and its Pareto frontier.** Each marker is one of 5,052 candidates evaluated across all six search runs (random + GA + four single-objective ablations) on the 66-dim keep-mask design space; colour encodes draw rate. The crimson line traces the actual 2D Pareto frontier of (fairness, rounds), both minimised: 14 points at 2p, 16 at 3p, with a clean knee around fairness 0.22 and rounds $30-40$ at 2p (a slightly higher knee at 3p). The gold star is the combined-objective GA optimum; it sits *inside* the cloud rather than on the 2D frontier because the composite weights five terms (fairness, max-fairness, length, draw, transfer), not just two; the GA's optimum is on the 5D Pareto surface, of which the (fairness, rounds) projection is a shadow. The 2p sweep reaches mean rounds as low as ~ 22 (the keep-mask shrinks the lap aggressively), while the 3p frontier bottoms out closer to ~ 35 rounds; the two clouds have visibly different shapes, which is what makes the cross-evaluation finding (Table 1) a property of the design space rather than seed noise.

A.6 LLM-driven GA convergence

The LLM-driven GA loop (Section 4.6) converges over its 32 evaluations and produces a winning board that beats the default under all-LLM evaluation; whether that board is a real design optimum or only an LLM-evaluator optimum is what Figure 8 resolves in the main text. The convergence curves and per-generation score distribution are below.



(a) LLM-GA best-so-far



(b) Score distribution

Figure 14: **An LLM-only evaluator can drive the same GA loop. (a)** Per-evaluation composite score (grey) and best-so-far (blue) over the GA's 32 evaluations (population 8, 5 generations, n_{seeds}=5, with elitism keeping the unique evaluation count at 32). The dashed reference is the rule-based GA-2p winner's score under the same all-LLM evaluator (0.728): the LLM-driven loop finds a board (red star, iter 15, score 0.465) that beats even the rule-based winner when scored by an LLM. **(b)** Per-generation distribution of candidate scores; the median visibly drops from generation 0 to generation 2 (selection pressure), and the spread within each generation reflects the n_{seeds}=5 evaluation noise. Total wall time 4h 58min on a single 12 GB GPU.

A.7 Training-vs-deployment overfit

A secondary finding worth recording: *optimisation-set overfit is small on both designs*. At the n=200 optimisation budget the GA-2p winner scored 0.705 on its training matchup set; at n=1000 deployment it scores 0.773, an overfit of +0.068 (Table 3). The GA-3p winner scored 0.599 at training and 0.641 at deployment, an overfit of +0.042. Both gaps are small ($\leq$ 9% of the deployment score) relative to the headline ∼47% improvement over the default, so the directional improvement is robust to re-evaluation at higher n.

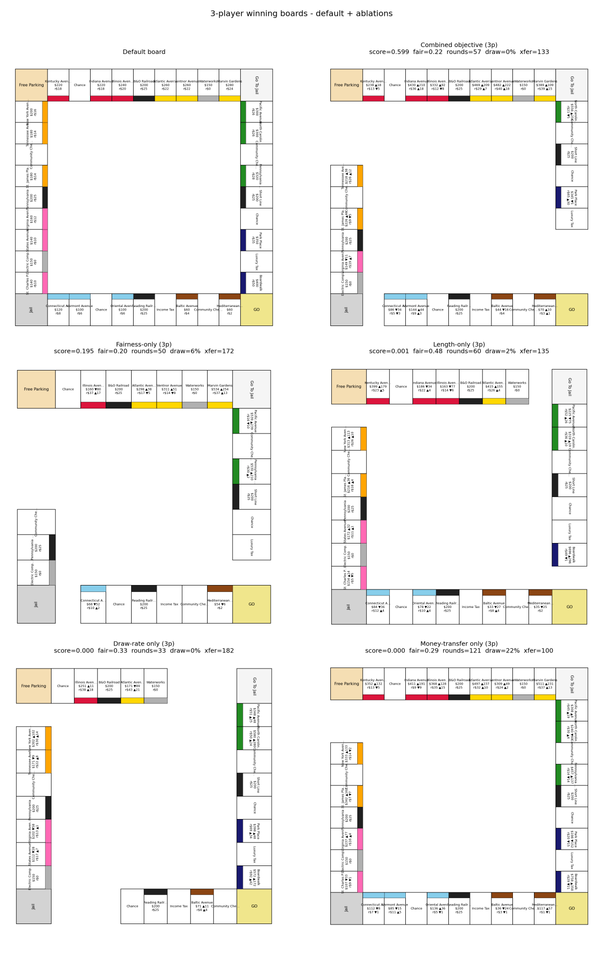
Design	training (n=200) ↓	deployment (n=1000) ↓	overfit ↓
GA-2p winner	0.705	0.773	+0.068
GA-3p winner	0.599	0.641	+0.042

Table 3: **Optimisation-set vs. deployment-set composite score, on each design's own training regime.** Header arrows: ↓ means lower is better. Both designs show small positive overfit (+0.068 and +0.042 respectively, both < 10% of the deployment score) relative to the headline ∼47% improvement over the default. Training number is the best score in the GA's own `evals.jsonl`; deployment number is the corresponding row of Table 1.

A.8 Single-objective ablation boards

The same non-redundancy claim from Section 4.2, made spatial: each ablation winner is a visibly different physical board, with different properties kept or dropped and different per-property cost / rent multipliers.

(a) 2 players



(b) 3 players

Figure 15: **Each ablation produces a visibly different physical board.** Six panels per player count: default (top-left) and the five GA-run winners (combined, fair-only, length-only, draw-only, money-only). Same non-redundancy finding as Figure 4, made spatial.

A.9 Evaluation methodology

Three standard techniques underpin the evaluation. Variance reduction via shared random numbers is standard in stochastic simulation [4]; Wilson confidence intervals [11] are the standard way to quote per-cell uncertainty on win rates; and per-objective single-knob ablations are standard in optimisation. We adopt all three because they are cheap, reproducible, and orthogonal; the only novelty is connecting them to the agent-as-design-probe framing.

A.10 GA, ablations, and reproducibility

Search hyperparameters (canonical protocol, logs in `logs/optimizer_v3/`).

- GA: `--pop 30 --generations 30 --elitism 2`, tournament selection k=3, uniform crossover, Gaussian mutation σ=0.1 on continuous dims, 1-bit-flip mutation on the keep-mask. Total evaluations: $30+29\times28 = 842$.
- Random search: `--iters 842` for matched evaluation budget.
- Per-candidate evaluation: `--n-games 200`, distributed as 10 fixed matchups × 20 games per matchup, with seeds shared across all candidates.
- Composite weights $w = (1, 0.5, 0.5, 0.3, 0.3)$ for $(\overline{F}, F_{\max}, |\overline{R}-R^*|/R^*, \overline{D}, |\overline{T}-T^*|/T^*)$ with targets $R^* = 60$ rounds, $T^* = 100$ \$/round.
- Single-objective ablations set the chosen weight to 1 and the others to 0; same search hyperparameters otherwise.
- Cross-evaluation at n=1000 games per cell (deployment table) uses the same matchup seeds.

Reproducibility.

- All seeds (`--base-seed=42`, `--search-seed=0`, `--matchup-seed=1234`) are recorded in each run's <run_name>.meta.json.
- Re-running any `(design vector, matchups, seeds)` triple yields byte-identical metrics. We verified this by diffing two re-runs of the same config.
- Cross-player-count scaling table data (Table 1) is in `logs/optimizer_v3/cross_eval_scaling.json`; the 2p / 3p columns reproduce the original `logs/optimizer_v3/cross_eval_mask.json` bit-for-bit.
- Trigger scripts: `scripts/rerun_strategy_experiments_overnight.bat` (rule-based, ∼2.5h CPU) and `scripts/rerun_llm_phase_c_v3.bat` (LLM-only evaluation re-run, ∼6h on a 12 GB GPU).

A.11 Limitations

- **Strategy-pool coverage.** The 17-dim parametric space is broad but not exhaustive, and human players behave outside it. The 20 sampled strategies broaden the pool beyond the 10 archetypes and the sampling procedure is documented and reproducible, but a real deployment would want to learn the strategy distribution from logs.
- **Human playtest is a small pilot, not a full study.** Only an N=10 two-player pilot was run (Section 4.7): two players played the default and the GA-2p combined winner under ten matched seeds, which confirms all four trained effects (pacing, money circulation, decisiveness, and win-balance) on this single subject pair but cannot rule out idiosyncrasy in playing style across the broader population. The full multi-subject playtest remains deferred; the shortlist in Appendix A.12 curates 4 boards and names the design hypothesis each is intended to probe, with a proposed pilot protocol (N=8, counterbalanced, Likert plus forced-choice).
- **Fairness is strongly dependent on the number of players.** The pacing, decisiveness, and composite advantages generalise cleanly from the 2p / 3p training regime to every count from 2 to 8 (Section 4.3, Figure 5), but fairness does not: the GA winners are 30−53% fairer than the default at training counts and 69−101% *less* fair at 7−8p. A deployment at a player count outside the training regime should re-optimise the fairness term against that count, or accept that fairness gains may not carry over.

- **LLM scale.** Qwen2.5-1.5B is the only LLM in the study. Larger models or different prompt styles may change the agreement profile. Greedy decoding does, however, make the probe deterministic.
- **Single-board scope.** All optimisation runs on the standard 40-cell US Monopoly layout (with structural shrinkage via keep-mask). Other game families would require a new design-space encoder, but the rest of the pipeline (objective, pool, GA, validator) carries over unchanged.

A.12 Full playtest (deferred): board shortlist

Beyond the N=10 pilot reported in Section 4.7, we did not run the full multi-subject playtest within the project window. Rather than report a synthetic surrogate, we publish the shortlist of boards we would playtest, with the design hypothesis each board is intended to probe. The shortlist is small (4 boards) so a pilot is feasible in a single 60–90 min session, and each board contrasts a different design lever against the default, so that even a handful of subjects gives readable signal on which axis humans actually respond to.

Boards.

1. **Default Monopoly** (control). Unperturbed `settings.py` board. Baseline for every subjective rating.
2. **GA-2p combined winner** (Section 4.1). Best composite-score 2p board: shorter, lower draw rate, and more inter-player money transfer than default. Tests *whether the combined objective tracks perceived game quality*, the core falsifiable claim of the GA half of the project.
3. **GA-3p combined winner** (Section 4.3). Cross-evaluation (Table 1) shows the 2p and 3p winners are structurally different boards; this comparison tests *whether human players can also feel the 2p$\not\leftrightarrow$3p design split*, or whether the difference is only legible to the rule-based pool.
4. **Money-only 2p optimum** (Section 4.2). Board produced by setting w_{money}=1 and zeroing every other weight, pushing per-round inter-player cash transfer from $\approx$ 50 \$/r (default) toward the target 100 \$/r while making no claim about fairness or pacing. Tests *whether interactivity* per se*, independent of fairness and length, drives perceived engagement.* This is the most novel of the four boards, because money-transfer rate is the only objective term the project introduces that has no direct precedent in the search-based game-design literature cited in Section 2.

Why these four, and not the other ablation winners. The shortlist deliberately excludes *length-only*, *draw-only*, and *fairness-only* optima. Length and draw rate also change in the combined-objective winner (board 2), so a playtest on those single-axis boards would not separate the hypothesis "length matters" from "the combined objective matters." Fairness is a property of the win-rate distribution across many games and is not directly observable from a single session: a single playtest game tells a subject who happened to win, not whether the board is fair. Money-only, by contrast, manipulates an axis that should be *felt within a single game* (more trades, more rent events, more state change per turn), so it is the only ablation winner where a small pilot would carry useful signal.

What we would measure. For each board: post-game Likert ratings (1–5) on pacing, fairness, engagement, and overall enjoyment, plus a forced-choice "would you play this version again." Between-board comparisons would be non-parametric (Wilcoxon signed-rank, Bonferroni-corrected across the 6 board pairs); per-board scores reported with bootstrap 95% CIs. The minimum useful pilot is N=8 subjects, each playing all four boards in counterbalanced order (Latin square on the 2p boards; the 3p board played in groups of three).

What this shortlist would falsify. The two strongest claims the GA-side of the report makes are (a) the combined objective produces a measurably different board, and (b) the per-objective ablations expose orthogonal design axes. A playtest where board 2 feels the same as board 1, or board 4 scores the same as board 1 on engagement, would mean the matching claim above is wrong. We frame the shortlist as a *falsification plan*, not a confirmation plan.

B Mapping to the CS348K grading questions

The CS348K final-report guidelines ask two specific questions. We state both verbatim and answer each in one paragraph; the body of the report is structured around exactly these two questions, so this appendix is a quick reader guide.

(1) "What are the questions or goals your project aims to answer?" We ask whether a diverse agent pool can serve as a beta-testing surrogate for human playtesting in game design under two qualifications: (a) the agents are imperfect and acknowledged as such, and (b) the unit of evidence is *cross-class agreement*: two independent agent classes (a 30-strategy parametric rule-based pool and a single-personality LLM with structured state validation) ranking the same candidate boards in the same direction. The falsifiable hypothesis is two-fold: (a) the directional ranking under the rule-based pool matches the directional ranking under the LLM seats on the same boards; and (b) the same directional ranking also holds in real human play, so the pool acts as a proxy for human behaviour on the design axes the optimiser targets. An N=10 two-player playtest pilot (Section 4.7) is the empirical check on (b); a fuller multi-subject playtest is published as a falsification plan (Appendix A.12). The full statement is in Section 1.

(2) "What experiments should be done to answer that question, and how will you know from the outcome of the experiment that you have succeeded?" Nine experiments, each with an explicit success criterion:

1. Default-board baseline at n=1000 (Section 4.1). *Success:* deterministic scores with 95% Wilson intervals on every metric.
2. GA vs. random search at matched ∼842-evaluation budget (Section 4.1). *Success:* GA crosses the default reference within budget, beats random by a non-trivial margin, and is bit-reproducible. *Outcome:* GA wins by 13% in both player counts.
3. Single-objective ablations (Section 4.2). *Success:* visibly different per-aspect optima, confirming the composite is non-redundant. *Outcome:* confirmed; see Figures 4 and 15.
4. $2p \leftrightarrow 3p$ cross-evaluation at n=1000 (Section 4.3). *Success:* improvement over default $\geq 20\%$ in the off-regime; quantify the specialisation gap. *Outcome:* both designs improve over default by $\sim 47\%$ in the training regime; off-regime gap is $16-24\%$ on the composite.
5. Archetype WR reshuffle (Section 4.4). *Success:* the GA winner produces an interpretable, non-trivial change in which named archetypes are top-tier vs. middling vs. bottom-tier, with the systematic pattern of which matchups are immutable surfaced honestly. *Outcome:* the GA winner reshuffles middle-of-distribution archetypes (Δ up to ± 0.18 in win-rate vs. the field, e.g. CashHoarder +0.18, HighCostOnly -0.18); the top-3 default strategies stay top-3 on the GA winner; *Trader vs. RailroadKing* (100/0) is genuinely immutable under any board configuration in the search space.
6. LLM cross-class probe (Section 4.5). *Success:* rounds and transfer move in the same direction the rule-based pool predicted on the same boards; per-decision hallucination rate below 1%. *Outcome:* 0/2,207 first-pass hallucinations under v2 ECHO validation; direction agrees on every measured metric at both 2p and 3p (e.g. rounds ∼−40%, transfer per player-turn ∼+60% at 2p).
7. LLM-driven GA + cross-class verdict (Section 4.6). *Success:* the LLM-driven loop converges to a non-trivial design; cross-evaluating it against the rule-based GA winner under both evaluator classes resolves which is the better design. *Outcome:* both evaluator classes prefer the rule-based GA winner on 4 of 5 stable metrics (composite, $\overline{F}$, $F_{\max}$, length) at *both* 2p and 3p (n=100 LLM seats each); the LLM-driven board wins only on transfer per player-turn, the metric its n=5 training overfit to. This is the central empirical claim of the paper.
8. Human playtest pilot (Section 4.7). *Success:* the design constraints specified in the agent setting are satisfied in real human play, in the same direction and similar relative magnitude as under the agent evaluators, and agent feedback is conservative rather than optimistic. *Outcome:* N=10 two-player matched-seed pilot; ∼50 vs. 131 rounds, \$66 vs. \$39 per round, 0% vs. 40% draws on the GA winner vs.

the default, with humans showing a *larger* default-board failure than either simulator (40% draws vs. 0–7% in pool/LLM). The agent-driven predictions are conservative rather than optimistic.

9. Playtest shortlist (Appendix A.12). *Success:* a small, falsifiable selection of boards from the GA shortlist, each contrasting a different design lever against the default, with a stated pilot protocol. *Outcome:* 4 boards selected (default, GA-2p combined winner, GA-3p combined winner, money-only 2p optimum) with one-sentence design hypotheses and a proposed pilot (N=8, counterbalanced, Likert plus forced-choice). The fuller multi-subject playtest is published as a falsification plan, not a completed study.

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